# Galaba Vamsi

Software Engineer | AI/ML | Computer Vision | Systems
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## PROFESSIONAL SUMMARY

Mechatronics undergraduate at IIT Bhilai focused on software engineering, AI/ML, computer vision, robotics, and GPU systems. Builds full-stack products, open-source tooling, simulators, and visual applications with Python, TypeScript, React, PyTorch, CUDA, and cloud platforms.

## EXPERIENCE

**Open-Source Contributor - HFlow** 2026
Hebbian Robotics (YC S26) | HFlow repository
- Delivered 6 maintainer-reviewed merged PRs addressing 6 tracked issues in HFlow's robotics and Physical AI SDK.
- PR #350 / issue #296: implemented paginated LeRobot tree traversal using `Link: rel="next"` headers, preserved request headers, deduplicated paths, rejected unsafe cross-origin URLs, and prevented pagination loops.
- PR #353 / issue #292: resolved LeRobot video-cache collisions by including camera, chunk, and file indexes in cache identity and bumping the converter version.
- PR #360 / issue #314: centralized safe ingest URI parsing across the CLI, server, and SDK and added validation for blank, absolute, and parent-escaping paths.
- PR #367 / issue #365: benchmarked 3 cold 1080p30 checks (900 frames/check; 4.897 s median) and separated transform timing/provenance; PR #400 / issue #399: refused LeRobot depth videos before conversion to preserve 12-bit depth semantics.

**Research Intern - 6G RIS Simulator** May 2025 - Aug 2025
IIT Bhilai | Advisor: Dr. Arzad Alam
- Built a zero-budget, open-source Python simulator with USRP-style interfaces for line-of-sight and non-line-of-sight wireless environments; RIM-SIM v2 includes 6 paper-grounded scenarios, a 20 Hz WebSocket feed, and 103 passing tests. [GitHub]
- Designed and validated a 2-layer reconfigurable intelligent surface (RIS) PCB workflow for self-powered RF energy harvesting; contributed to 3 related publications (1 as first author and 2 as co-author).

**Marketing Intern - Campus CEO Project** May 2024 - Jul 2024
Swiggy Ltd
- Managed a campaign with 15 social media posts; contributed to campus marketing initiatives and execution.

## PROJECTS

**Human Slop - Co-Founder** 2026 - Present
Built and deployed an anti-AI web and Android platform with React, PostgreSQL, and AWS; implemented hardware-bound biometric authentication and real-time typing forensics. [Website] | [App] | [GitHub]

**Arista Networks Wi-Fi Optimization System - Inter IIT 5th Rank** 2025
Built a client-aware Wi-Fi RRM system in NS-3 using 3 RF metrics; classified 100,000 I/Q samples from 5 ms captures into 16 classes with STFT, DBSCAN, and Dual-CNN inference; simulated 5 access points and 50 client stations using IEEE 802.11k/v. [Slides]

**Electron-GNN - ML-Accelerated Quantum Spectroscopy** 2026
Built a V3 two-tower GATv2 model to predict molecular absorption spectra from geometry; evaluated on 2 molecules with 39 and 55 target peaks. [GitHub]. Faculty advisor: Dr. Soumajit Pramanik.

**Open-PyFX - GPU-Accelerated Video Effects** 2025
Built a Python and GLSL video effects suite with GPU rendering for CRT emulation, ASCII edge detection, retro graphics, and 3D LUT pipelines; supports 3840x2160 export and up to 16x zoom. [GitHub]

**Lunar Crater Detection - DEM Terrain Analysis** 2025
Built a computer vision pipeline for crater extraction from Digital Elevation Models; achieved an F1 score of 0.839 and approximately 4x preprocessing speedup using CuPy/CUDA; the repository includes 77 tests. [GitHub]

**AntennaNet - Inverse Electromagnetic Design** 2025
Developed KD-tree nearest-neighbour spectral anchoring for inverse electromagnetic design across 144D search spaces using Python, MEEP, and OpenMPI; indexed approximately 2,000 spectra and a 1,200-sample, 12x12-pixel dataset. [GitHub]

**Amazon ML Challenge - Top 3.5%** 2025
Multimodal price prediction with Qwen 2-VL 7B and a ViT/RoBERTa two-tower model; 75,000 data pairs; SMAPE 49.0.

## SKILLS

**Languages:** Python, C/C++, TypeScript, JavaScript, SQL, GLSL, CUDA.
**AI/ML:** PyTorch, Graph Neural Networks (GATv2), Safe Reinforcement Learning, Transformers, OpenCV, CuPy, NumPy.
**Software and Systems:** React, React Native, Flutter, FastAPI, Node.js, REST APIs, WebSockets, PostgreSQL, AWS, Supabase, Docker, ROS 2, Gazebo, MEEP, OpenMPI, SDR/USRP.
**Developer Tools:** Git, Linux, GitHub Actions, CI/CD, pytest, Ruff, ty, static analysis, regression testing.

## EDUCATION

**B.Tech. in Mechatronics** Aug 2023 - May 2027
Indian Institute of Technology Bhilai
**Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Deep Learning for Computer Vision, AI and Machine Learning, Digital Image Processing, Optimal Control.

## PUBLICATIONS

**First author:** *An Open Emulator for Smart Radio Environments*. [Link]
**Co-author:** *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025. [Link]
**Co-author:** *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*. [Link]

## ACHIEVEMENTS

**1st Place:** IIT Bhilai Web Development Competition.
**5th Rank:** Arista Networks Wi-Fi Challenge, Inter IIT Tech Meet 14.0.
**Top 3.5%:** Amazon ML Challenge.
**YC Startup School:** Human Slop accepted.
**Top 10 Finalist:** Toyota Hackathon.